Normal Distribution

Recall

For a continuous random variable X, with pdf $f(x)$ and cdf $F(x)$ the followings were true.

$$F(x) = \int_{-\infty}^{x} f(t)dt \tag{1}$$

$$P(a \leq X \leq b) = \int_{a}^{b} f(t)dt = F(b) - F(a) \tag{2}$$

$$E[X] = \int_{-\infty}^{\infty} t \cdot f(t)dt \tag{3}$$

- Normal distribution is the most useful distribution. Normal distribution is also known as the Gaussian distribution or the "bell curve."
- By a theorem that we will learn in early future, we will see that under a very weak assumption, almost everything has a normal distribution.
- For example we will see that for a very large n we can approximate a binomial distribution with normal distribution, and generally because of properties of normal distribution the probabilities are easier to calculate using normal distribution.

Definition (Normal Distribution $Y \sim N(\mu, \sigma^2)$)

Y has a normal distribution with mean μ and standard deviation σ if Pdf Y is given by

$$\begin{aligned} f_Y(y) &= \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(y-\mu)^2}{2\sigma^2}} \text{ where } -\infty < y < \infty \\ E[Y] &= \mu \\ Var(Y) &= \sigma^2 \end{aligned}$$

Definition ($Z \sim N(0,1)$)

The normal distribution with $\mu = 0$ and $\sigma = 1$ is called the standard normal distribution. The random variable with Standard normal distribution is shown with Z. The pdf in this case is shown by $\varphi(z)$ and the cdf is shown by $\Phi(x)$

$$\varphi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{z^2}{2}} \text{ where } -\infty < z < \infty$$

$$\Phi(x) = P(Z \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{z^2}{2}}\, dz$$

Note: Generally it is not possible to calculate the above integral by basic calculus method, we should use numerical methods and software. So what to do if we want to calculate $P(a < Z < b)$? We know that $P(a < Z < b) = \Phi(b) - \Phi(a)$, if there is no computer available, we will use the Standard normal CDf table to calculate this probability.

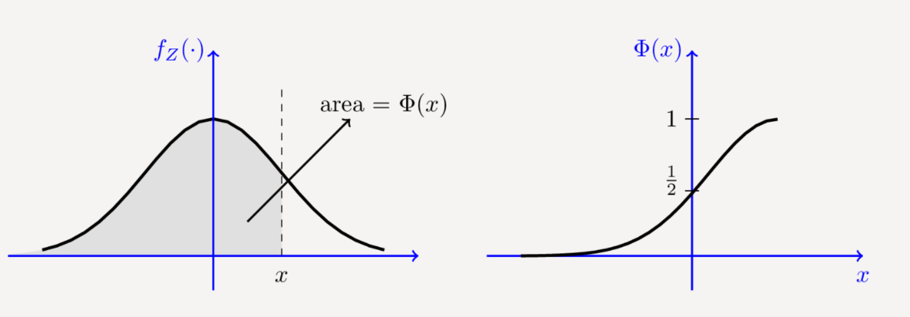


Figure: Graph of pdf of standard normal distribution. The Pdf is symmetric with respect to mean.

Properties of CDF of a Standard Normal

(1) $\Phi(x) = 1 - \Phi(-x)$, for every $x \in \mathbb{R}$

(2) $\Phi(0) = \dfrac{1}{2}$

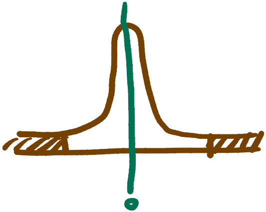
(3) Similar to all the other CDF functions $\lim_{x \to -\infty} \Phi(x) = 0$, $\lim_{x \to \infty} \Phi(x) = 1$.

Eg 1. Find $P(0 \leq Z < 1.83)$

$$P(0 \leq Z < 1.83) = \Phi(1.83) - \Phi(0)$$

$$= 0.9664 - 0.5$$

$$= 0.4664$$



Eg 2. Find $P(Z > 2)$

We will use symmetry to solve it.

$$P(Z>2) = P(Z<-2) = P(Z \leq -2) = \Phi(-2)$$

continuous

Proposition

Let $X \sim N(\mu, \sigma^2)$,

(1) $Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$.

(2) $F(x) = \Phi(Z) = \Phi(\dfrac{X-\mu}{\sigma})$.

(3) $f(x) = \dfrac{1}{\sigma}\varphi(\dfrac{X-\mu}{\sigma})$

Eg 2. Let $X \sim N(\overset{\mu}{-5}, \overset{\sigma^2}{9})$ $\Rightarrow$ $Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$

$Z = \dfrac{X+5}{3} \sim N(0,1)$

(a) Find $P(X < 0)$.

(b) Find $P(-6 < X < -2)$.

(a) $P(X<0) = P(X-(-5) < 5)$

$= P\left(\dfrac{X+5}{3} < 5/3\right) = P(Z < 5/3)$

$= \Phi(5/3)$

(b) $P(-6 < X < -2) = P(-1 < X+5 < 3)$

$= P\left(-1/3 < \dfrac{X+5}{3} < 1\right)$

$= P(-1/3 < Z < 1)$

$= \Phi(1) - \Phi(-1/3)$

You must know the following rule by heart.

68-95-99 Rule:

Generally if $X \sim N(\mu, \sigma^2)$, then
(1) The probability of falling within 1 standard deviation of the mean is about 0.68, i.e.

$$P(|X - \mu| < \sigma) \approx 0.68$$

(2) The probability of falling within 2 standard deviation of the mean is about 0.95, i.e.

$$P(|X - \mu| < 2\sigma) \approx 0.95$$

(3) The probability of falling within 3 standard deviation of the mean is about 0.997, i.e.

$$P(|X - \mu| < 3\sigma) \approx 0.997$$

Eg 3. Let $X \sim N(-1, 4)$. What is $P(|X| < 3)$? Do not use standard cdf table.

$$Z = \frac{X-\mu}{6}$$

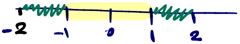
$$Z = \frac{X+1}{2} \sim N(0,1)$$

$$P(-3 < X < 3) = P(-2 < X+1 < 4)$$

$$= P\left(-1 < \frac{X+1}{2} < 2\right)$$

$$= P\left(\left|\frac{X-\mu}{6}\right| < 1\right) + P(1 \leqslant Z < 2)$$

$$= P(|Z| < 1) + P(1 \leqslant Z < 2)$$



$$P(|Z| < 1) = 0.68$$

$$P(|Z| < 2) = 0.95$$

$$P(1 \leqslant Z < 2) = \frac{0.95 - 0.68}{2} = \frac{0.27}{2} = 0.135$$

$$P(-3 < X < 3) = P(|Z| < 1 + P(1 \leqslant Z < 2)$$

$$= 0.68 + 0.135 = 0.815$$